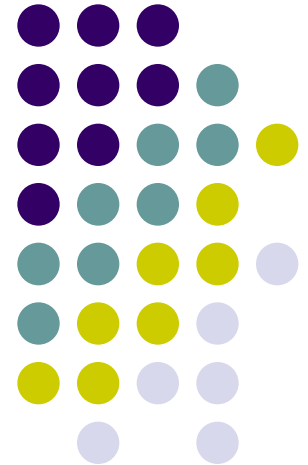
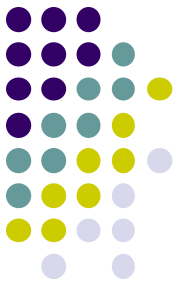


Natural Language Processing

CPSC 488

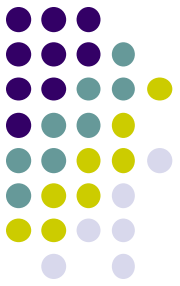
Distributional Vector Semantics



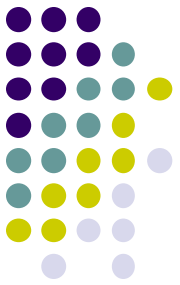


Lecture Overview

- Word semantics
 - Word meaning
 - Related words and word similarity
 - Distributional hypothesis
- Distributional vector semantics
 - Computational models of word meaning
- Word embedding and vectorization
 - Singular Valued Decomposition (SVD) for embedding
 - Brown clustering as a class-based word vectorization

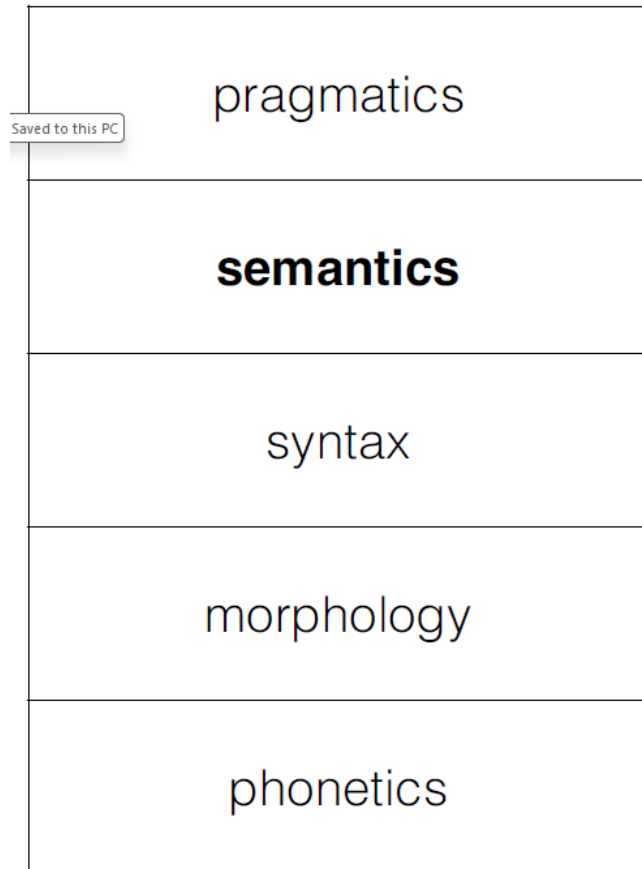


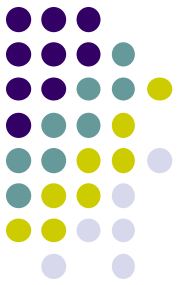
Word Semantics



A Linguistic Hierarchy

- Linguistic hierarchy
 - The structured arrangement of language element
- Pragmatics
 - Concerns how speakers imply meaning and how listeners infer it, taking into account factors like the speaker's intention, the context of the interaction, and social conventions.
- Semantics
 - Structure is important but one way as a “scaffolding for meaning”





Representing the Word Meaning

- **Word meaning** by Webster dictionary
 - The **idea** that is **represented** by a word, phrase, etc.
 - The **idea** that a person wants to **express** by using words, signs, etc.
- **Linguistic way of thinking**

signifier (symbol) $\Leftrightarrow$ signified (idea or thing)

= denotational semantics

tree $\Leftrightarrow$ {, , , ...}

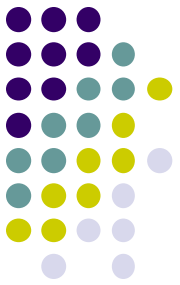
Language

Language and thought

Ludwig Wittgenstein (1889-1951):

“How can we know whether our ideas are clearly expressed in language?”

For clarity is not secured by the words themselves, but by their place in our shared practices.”



Representing the Word Meaning

- **Word meaning**

- The **idea** that is represented by a word, phrase, etc.
- The **idea** that a person wants to express by using words, signs, etc.

- **N-gram or text classification methods**

- **Words** are just **strings** (or **indices** w_i in a vocabulary list)
- This is not very satisfactory!

- **Logics**

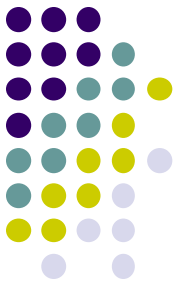
- The meaning of “**mouse**”: $\text{MOUSE}, \forall x \text{ MOUSE}(x) \rightarrow \text{MAMMAL}(x)$
- That seems hardly better!

- **Denotational and referential semantics**

- Meaning as the reference of expressions to things, e.g., $\text{Cat} \rightarrow \{\text{all cats}\}$
- Formal method and programming languages, e.g., $x + 2$ or $f(x) + b$ in a program
- Abstract meaning from mathematical objects!

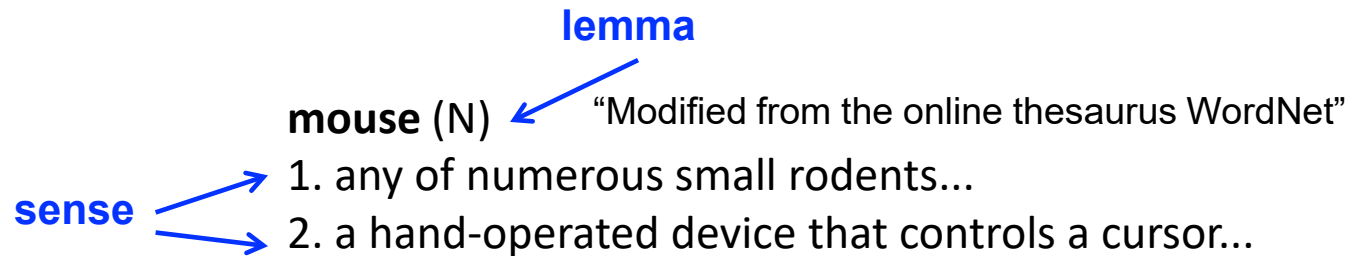
- **Lexical semantics**

- **What's the meaning of life? LIFE** (Carlson, 1977)
 - Surely, we can do better than this.
- **The linguistic study of word means as their relationships**



Lexical Semantics

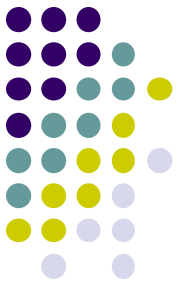
- **Word meaning as their relationships**



- Lemmas can be polysemous (having multiple senses).
 - "Sense" or "concept" is the meaning component of a word.

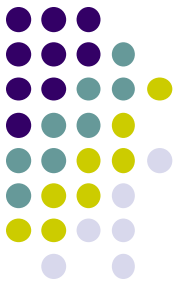
- **Linguistic representation of the word meaning**

- Words, definitions, lemmas, senses
- Relationships between words or senses
- Taxonomy relationships
- Word similarity, word relatedness
- Semantic frames, semantic roles
- Connotation and sentiment



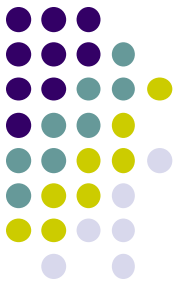
Relationships

- **Many-to-Many relationships** between **form** and **meaning**
- **Same forms**
 - Polysemy **related meanings**
 - Book (text) vs book (reservation), head (of a person) vs head (of a company)
 - Homonymy **different meanings**
 - Words spelled or pronounced the same but have different or unrelated meanings
 - Bat (animal) vs bat (baseball)
- **Different forms**
 - Synonymy **same/similar meanings**
 - Antonymy **opposite/contrary meanings**



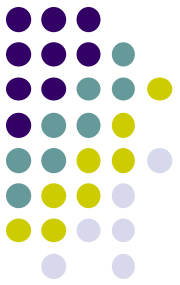
More Relationships

- Hypernym (general class or category)
 - IS-A (animal, cat)
 - Cat → feline → carnivore → placental mammal → mammal → vertebrate → ...
- Meronymy (part/whole)
 - HAS-PART (cat, paw)
 - IS-PART-OF (paw, cat)
- Membership
 - IS-MEMBER-OF (professor, faculty)
 - HAS-MEMBER (faculty, professor)



Relation: Synonymy

- **Synonyms** have the same meaning in some or all contexts.
 - couch / sofa
 - big / large
 - automobile / car
 - vomit / throw up
 - filbert / hazelnut
 - water / H₂O
- **Perfect synonymy?**
 - Probably no examples of perfect synonymy
 - Even if many aspects of meaning are identical, still may differ based on politeness, slang, register, genre, etc.
 - Consider big/large: my big sister != my large sister
- **The Linguistic Principle of Contrast**
 - Difference in form → Difference in meaning

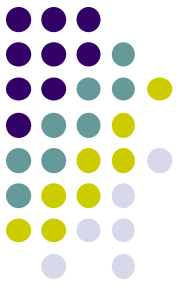


Relation: Antonymy

- **Opposite senses** with respect to **only one feature of meaning**. Otherwise, they are very **similar** or **related**!

dark/light short/long fast/slow rise/fall
hot/cold up/down in/out

- More formally: **antonyms can**
 - Define a binary opposition or be at opposite ends of a scale
 - long/short, fast/slow
 - Be reverse
 - rise/fall, up/down



Relation: Similarity

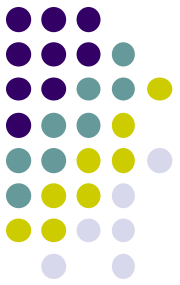
- **Words** with **similar meanings**, *not synonyms*, but **sharing** some element of meaning

- Car, bicycle
- Cow, horse

Asked **humans** how similar these two words are.

word1	word2	similarity
vanish	disappear	9.8
behave	obey	7.3
belief	impression	5.95
muscle	bone	3.65
modest	flexible	0.98
hole	agreement	0.3

SimLex-999 dataset (Hill et al., 2015)



Relation: Relatedness

- World **relatedness** or **association**

- Words can be **related** in any way, perhaps via a **semantic field**
 - coffee, tea: **similar**
 - coffee, cup: **related**, **not similar**

A semantic field is a set of words which cover a particular semantic domain and bear structured relations with each other.

- **Semantic field**

- A set of words or expressions that are related in meaning

hospitals

surgeon, scalpel, nurse, anaesthetic, hospital

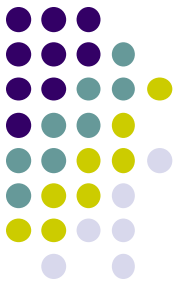
restaurants

waiter, menu, plate, food, menu, chef

houses

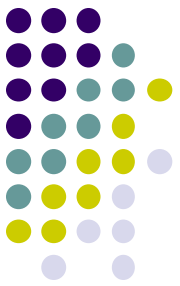
door, roof, kitchen, family, bed

- How to represent the semantic field? => **Semantic frame**



Semantic Frame

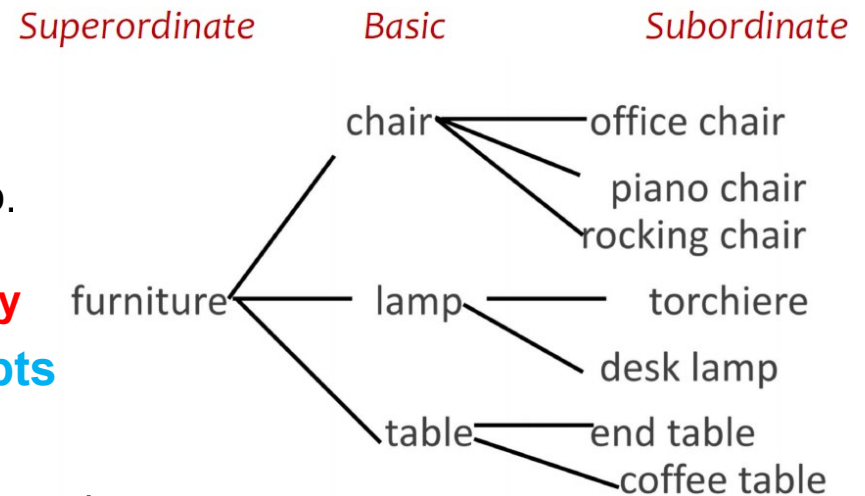
- A **cognitive structure** that organizes **knowledge** about a particular concept or situation
 - A set of **related meanings** and **associated roles** or **participants** involved in that **concept**.
 - A **set of words** that **denote perspectives** or **participants** in a particular type of event
 - “buy” (the event from the perspective of the buyer)
 - “sell” (from the perspective of the seller)
 - “pay” (focusing on the monetary aspect)
 - John hit Bill
 - Bill was hit by John
- **Frames** have **semantic roles** (like buyers, sellers, goods, money) and **words** in a sentence can take on those roles.



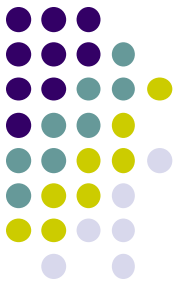
Relation: Superordinate/Subordinate

- One sense is a **subordinate** of another if the first sense is **more specific**, denoting a subclass of the other.
 - Car is a subordinate of vehicle.
 - Mango is a subordinate of fruit.
- **Conversely superordinate**
 - Vehicle is a superordinate of car.
 - Fruit is a superordinate of mango.

Taxonomy
A hierarchy of concepts



Object-Oriented concepts in software engineering are derived from this type of knowledge representation.



Connotation

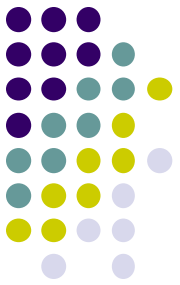
- **Connotation**

- **Implied meaning** of a word beyond its literal definition (denotation)
 - Nuance, emotional undertone, cultural or social associations
- **Connotations** can be **subtle**.
 - Positive connotation: *copy, replica, reproduction*
 - Negative connotation: *fake, knockoff, forgery*

- **Connotation in sentiment-bearing words**

- Words have **affective** meanings
 - Positive connotations (*happy*)
 - Negative connotations (*sad*)
- **Evaluation** – sentiment analysis
 - Positive evaluation (*great, love*)
 - Negative evaluation (*terrible, hate*)

- Word meaning in somewhere **in between?**



Connotation: Words in 3D Space

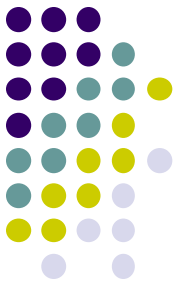
- **Words** seem to vary along **3 affective dimensions** (Osgood et al., 1957):
 - **Valence**: the degree of pleasantness of the stimulus (happy, hate)
 - **Arousal**: the intensity of emotion provoked by the stimulus (crazy)
 - **Dominance**: the degree of control exerted by the stimulus (powerful)

The meaning for each dimension

	Word	Score	Word	Score
Valence	love	1.000	toxic	0.008
	happy	1.000	nightmare	0.005
Arousal	elated	0.960	mellow	0.069
	frenzy	0.965	napping	0.046
Dominance	powerful	0.991	weak	0.045
	leadership	0.983	empty	0.081

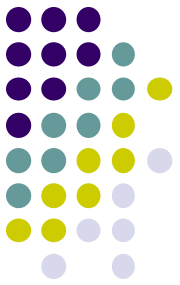
Dimension vectors

	Valence	Arousal	Dominance
courageous	8.05	5.5	7.38
music	7.67	5.57	6.5
heartbreak	2.45	5.65	3.58
cub	6.71	3.95	4.24
life	6.68	5.59	5.89



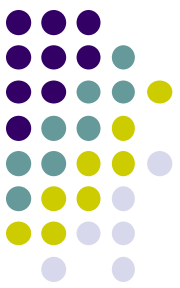
Characteristics of Word Meaning

- **Words have many meanings**
 - She pays 3% **interest** on the loan.
 - He showed a lot of **interest** in the painting.
 - It is in the national **interest** to invade the Bahamas.
- **Words are related and overlapped in meaning**
 - **What is the relationship among these words?**
 - {organization, team, group, association, consortium, federation, alliance}
 - Organization?
- **Word senses can be organized – relationships**
 - A group of words with a shared meaning, e.g. synonyms, antonyms (in Wordnet)
 - This generalizes the notion of a word.
 - We can then define relationships between these sets of words, semantic similarity, word sense disambiguation, ontology building, etc.
- **Words have implied meaning (connotation)**
 - Hard to express the subtlety: e.g., sentiment words meaning in 3D space



So far, we have discussed

- **Several ways to represent word meaning**
 - Symbolic notations, N-grams, Logics, Denotational or referential semantic
 - Lexical semantics – linguistic representation – word meaning as their relations
- **Words have relations**
 - Many-to-many relationships between form and meaning
 - Synonymy
 - Antonymy
 - Similarity
 - Relatedness, semantic field and frame
 - Word hierarchy (subordinate/superordinate)
 - Connotation (implied meaning) and its subtlety
 - The linguistic principle of contrast
- **The characteristics of word meaning**
 - Many meanings, Overlapped in meaning, Organized, related, and shared in meaning, words have implied meaning
- **Lexical semantics to represent the word meaning**
 - Word meaning as their relationships
 - Word meaning may be more accurately represented in 3D space.



Representing Word Meaning in Computers using WordNet in the past

- Previously the most common NLP solution:
 - Electronic dictionaries, e.g., **WordNet**, a thesaurus containing lists of synonym sets and hypernyms (“IS-A” relationships)

e.g., synonym sets containing “good”:

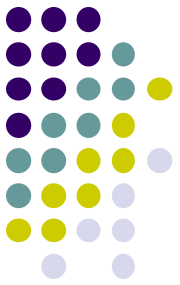
```
from nltk.corpus import wordnet as wn
poses = { 'n': 'noun', 'v': 'verb', 's': 'adj (s)', 'a': 'adj', 'r': 'adv' }
for synset in wn.synsets("good"):
    print("{}: {}".format(poses[synset.pos()],
        ", ".join([l.name() for l in synset.lemmas()])))
```

```
noun: good
noun: good, goodness
noun: good, goodness
noun: commodity, trade_good, good
adj: good
adj (sat): full, good
adj: good
adj (sat): estimable, good, honorable, respectable
adj (sat): beneficial, good
adj (sat): good
adj (sat): good, just, upright
...
adverb: well, good
adverb: thoroughly, soundly, good
```

e.g., hypernyms of “panda”:

```
from nltk.corpus import wordnet as wn
panda = wn.synset("panda.n.01")
hyper = lambda s: s.hypernyms()
list(panda.closure(hyper))
```

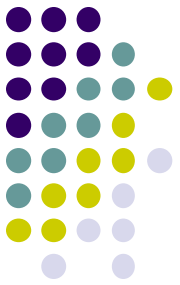
```
[Synset('procyonid.n.01'),
Synset('carnivore.n.01'),
Synset('placental.n.01'),
Synset('mammal.n.01'),
Synset('vertebrate.n.01'),
Synset('chordate.n.01'),
Synset('animal.n.01'),
Synset('organism.n.01'),
Synset('living_thing.n.01'),
Synset('whole.n.02'),
Synset('object.n.01'),
Synset('physical_entity.n.01'),
Synset('entity.n.01')]
```



Problems with Resources like WordNet

- **A useful resource but missing nuance**
 - Too coarse: “**proficient**” is listed as a synonym for “**good**”, expert → skillful
 - This is only correct in some contexts.
- **Missing new words**
 - e.g., **wicked**, **badass**, **nifty**, **wizard**, **genius**, **ninja**, **bombest**
 - Impossible to keep up-to-date!
- **Subjective**
 - WordNet list offensive synonyms in some synonym sets **without** any coverage of the **connotations** or appropriateness of words.
- **Expensive**
 - Requires human labor to create and adapt
- **Discrete representation**

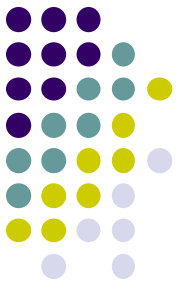
<i>expert</i>	[	0	0	0	1	0	0	0	0	0	0	0	0	0	0]
<i>skillful</i>	[	0	0	0	0	0	0	0	0	0	0	1	0	0	0]
- **Can't accurately compute word similarity or relationships**



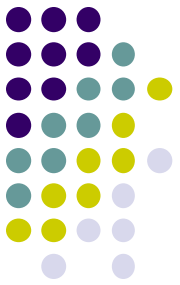
Better Models for Word Meaning

- **Two big ideas!**
 - **Idea 1:** Defining meaning using **linguistic distribution**
 - **Idea 2:** Meaning as **a point in multidimensional space**
- We'll introduce **vector semantics** based on these ideas.
 - Computational models for word meaning
 - Achieves many of our NLP goals!
 - Now, the standard model in NLP!

Idea 1: Meaning by Linguistic Distribution



- Ludwig Wittgenstein (Austrian philosopher, 1889):
 - "The meaning of a word is its **use** in the language, also implying the limitation of a language"
- Zellig Harris (structural linguistics, 1934, professor of Noam Chomsky):
 - If A and B have almost identical **environments**, we say that they are synonyms.
 - Words are defined by **their environments** (the words around them).
- Firth (linguist, 1957)
 - "You shall know a word by the **company it keeps**"
- **One way to define "usage"**
 - Its neighbor words or grammatical environments
 - The meaning of a word is defined by its distribution in **language use**.



Idea 2: Meaning as a Point in Space

- **Example: Three affective dimensions for a word**

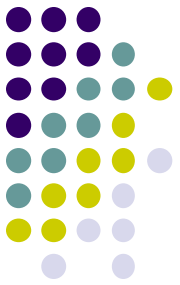
(Semantic differential, Osgood et al., 1957, Psycholinguistics)

- **valence**: pleasantness
- **arousal**: intensity of emotion
- **dominance**: the degree of control exerted

NRC VAD Lexicon (Mohammad 2018)

	Word	Score		Word	Score
Valence	love	1.000		toxic	0.008
	happy	1.000		nightmare	0.005
Arousal	elated	0.960		mellow	0.069
	frenzy	0.965		napping	0.046
Dominance	powerful	0.991		weak	0.045
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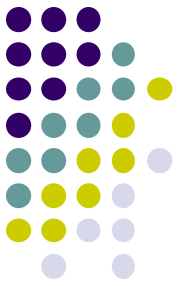
- Hence, the **connotation of a word** is **a vector in 3D space**.
 - **Meaning as a point in multidimensional space**



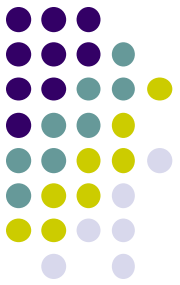
Meaning as a Point in Vector Space

- **We define a meaning of a word as a vector.**
 - Each word represented as a vector
- **Vector semantics:**
 - **Similar words** are "**nearby in semantic space**".
 - We build this vector space by seeing **which words are nearby** in text.





Computational Models of Word Meaning



Discrete Representation of Words

- In traditional NLP, we regard words as **discrete symbols**: *hotel*, *conference*, *motel* (a localist representation)

Means one 1, the rest 0s

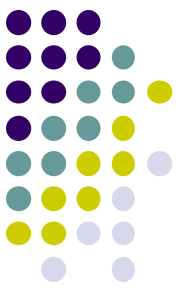


symbols for words can be represented by **one-hot** vectors:

motel = [0 0 0 0 0 0 0 0 0 1 0 0 0 0]

hotel = [0 0 0 0 0 0 0 1 0 0 0 0 0 0]

- **Vector dimension** = **number of words** in vocabulary (e.g., 500,000+)
- **Problems with words as discrete symbols**
 - If a user searches for “*Seattle motel*”, we would like to match documents containing “*Seattle hotel*”.
 - But these two vectors are orthogonal, **no natural notion of similarity**.
- **Possible solutions**
 - *WordNet*’s list of synonyms to get similarity, but it is well-known to fail badly due to incompleteness, subjectivity, etc.
 - **Machine learning to learn to encode similarities** in the **vectors**.



Distributional Semantics

- **Based on distributional hypothesis**

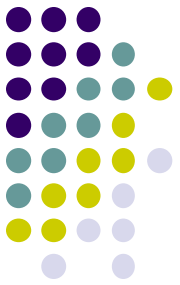
- “The **meaning** of a word is **its use** in the language” (Wittgenstein, 1900s)
- “If A and B have **almost identical environments**, we say that they are synonyms” (Harris, 1934, linguist)
- “You shall know a word by the **company it keeps**” (Firth 1957, linguist)

- **Contextual representation of words**

- When a word **w** appears in a text, its **context** is the set of words that **appear nearby** called “context words” (typically **within a fixed-size window**).
- The **meaning** and **its representation** of word **w** based on its **context words**.

...government debt problems turning into **banking** crises as happened in 2009...
...saying that Europe needs unified **banking** regulation to replace the hodgepodge...
...India has just given its **banking** system a shot in the arm...

↑ ↑
These **context words** will represent **banking**



Distributional Vector Semantics

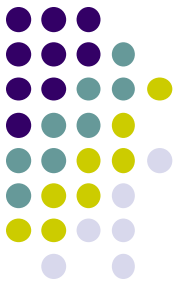
- **Distributional semantics**

- A word's meaning is given by the words that frequently appear close-by (context words).
- The word meaning is **defined** and **represented** based on its context.

- **Vector semantics**

- “Similar words are **nearby in semantic space**”.
- “Words that occur in **similar contexts** tend to have **similar meanings**”.
- The word meaning is defined and represented in a **vector space**.
 - What's the big deal about the vector representation of word meaning?

- **One of the great ideas of modern statistical NLP!**



Various Word Vector Representation

- **Count-based**

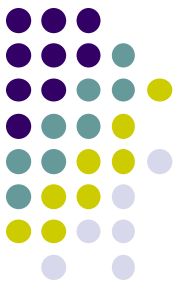
- Document-Term or Term-Document Matrix, Word-Context Matrix
- TF-IDF
- Pointwise Mutual Information (PMI), Positive PMI (PPMI)

- **Class-based**

- Brown clustering algorithm

- **Embeddings** (for dense vectors)

- Singular Value Decomposition (SVD) – static
- Distributed prediction-based embeddings – static
 - Word2vec, FastText
- Distributed contextual embeddings from language models – dynamic
 - Elmo, BERT
- Many more variants
 - Multilingual embeddings, multi-sense embeddings, syntactic embeddings, etc.



Vectors from Term-Document Matrix

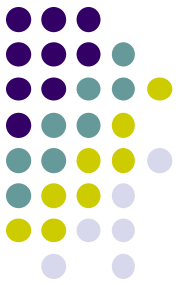
	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

Transpose of
Document-Term Matrix,
 DTM^T

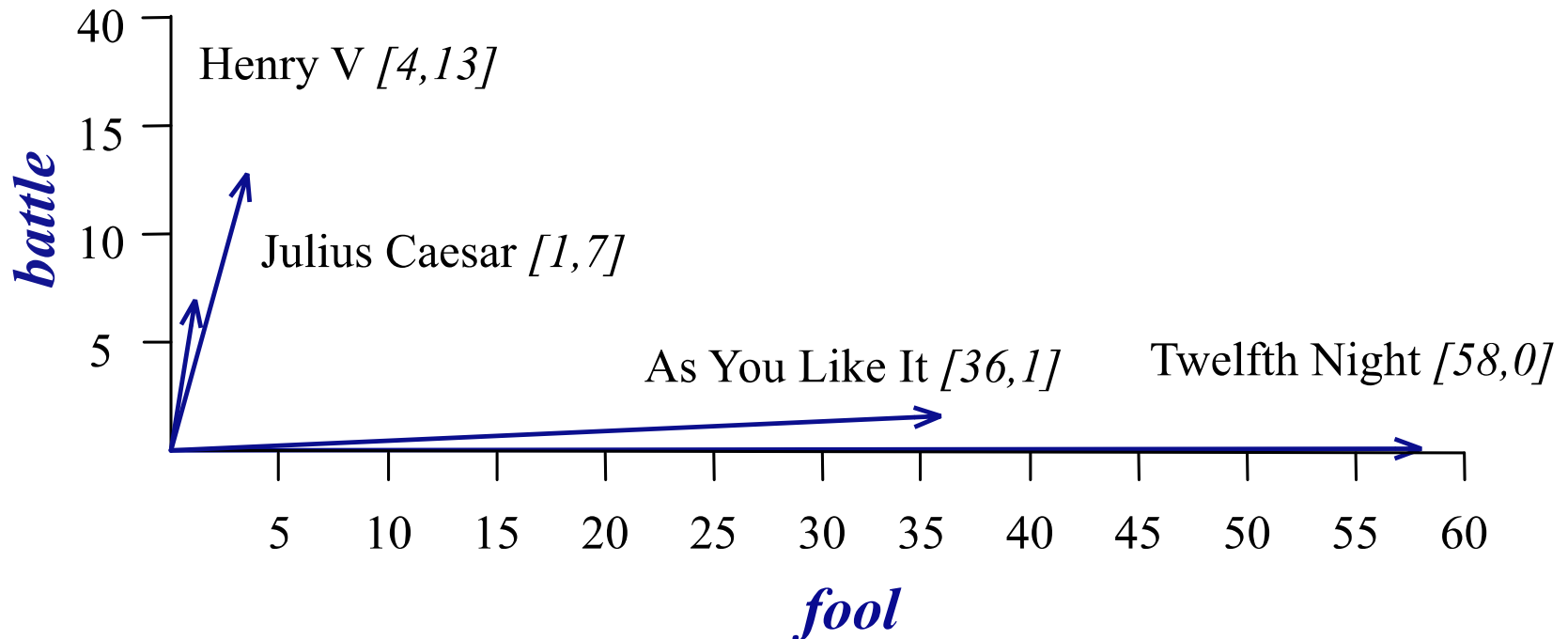
- A **document** as a vector of words
 - A compressed representation (encoding) of the document's content
 - When **two documents are similar**, the **corresponding vectors are similar**.

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

- A **word** as a vector of documents
 - A word vector encodes its document distribution
 - **A profile of documents, or possible word meaning**
 - The word **“battle”** occurs in “Julius Caesar” and “Henry V”
 - The word **“fool”** occurs in comedies, especially “Twelfth Night”

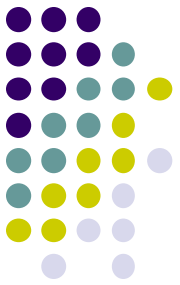


Visualizing Document Vectors



What can we do with the vectorized words?

Now we can perform the vector operations for text data!



Term-Context Matrix

(Term-Term or Word-Word Matrix)

Context words (columns): $P(w, c) = \text{raw count}$

Words
(rows)

	computer	data	result	pie	sugar	<u>count(w)</u>
cherry	2	8	9	442	25	486
strawberry	0	0	1	60	19	80
digital	1670	1683	85	5	4	3447
information	3325	3982	378	5	13	7703
<u>count(context)</u>	4997	5673	473	512	61	<u>11716</u>

A word as a vector of context words

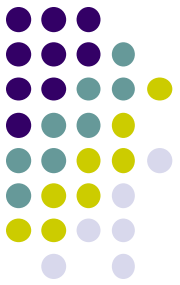
Two words are similar in meaning if their context vectors are similar.

Similarity==Relatedness

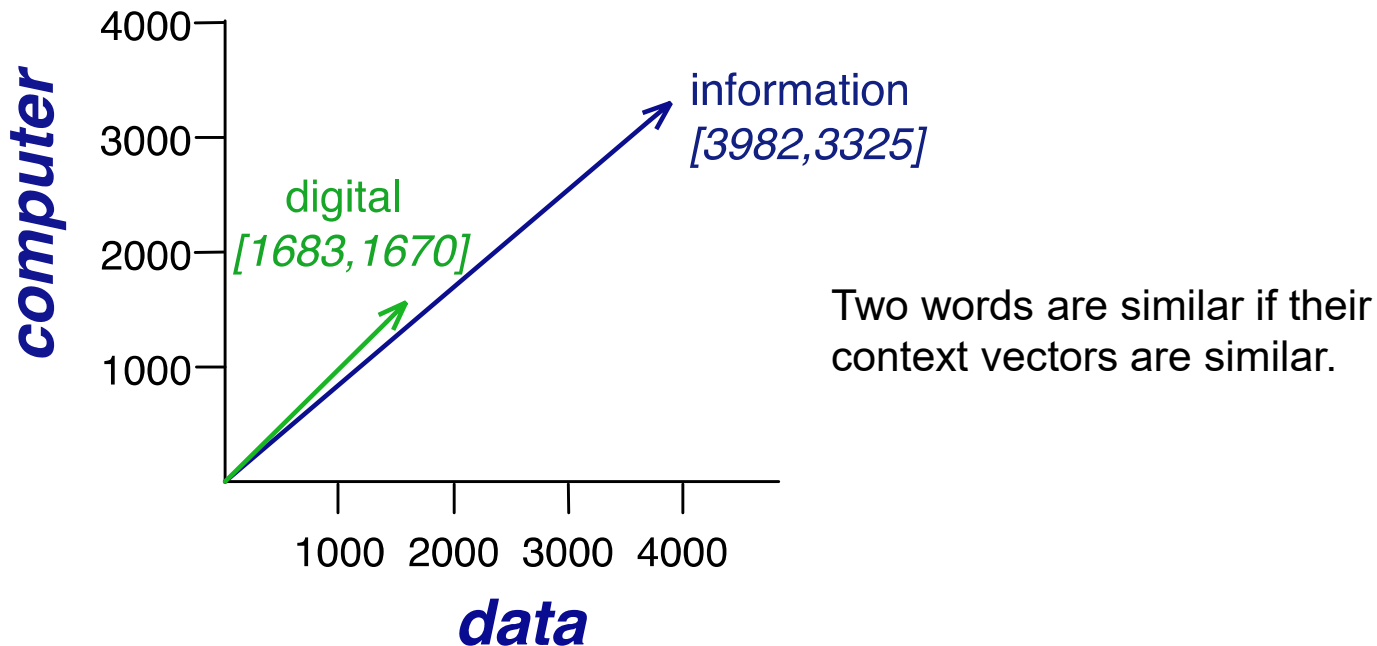
Word probabilities: $P(w, c) = \text{probability}$

	<u>p(w,context)</u>					<u>p(w)</u>
	computer	data	result	pie	sugar	<u>p(w)</u>
cherry	0.0002	0.0007	0.0008	0.0377	0.0021	0.0415
strawberry	0.0000	0.0000	0.0001	0.0051	0.0016	0.0068
digital	0.1425	0.1436	0.0073	0.0004	0.0003	0.2942
information	0.2838	0.3399	0.0323	0.0004	0.0011	0.6575
<u>p(context)</u>	0.4265	0.4842	0.0404	0.0437	0.0052	

- $P(\text{information, data}) = 3982/11716 = 0.3399$
- $P(\text{information}) = 7703/11716 = .6575$, $P(\text{data}) = 5673/11716 = .4842$

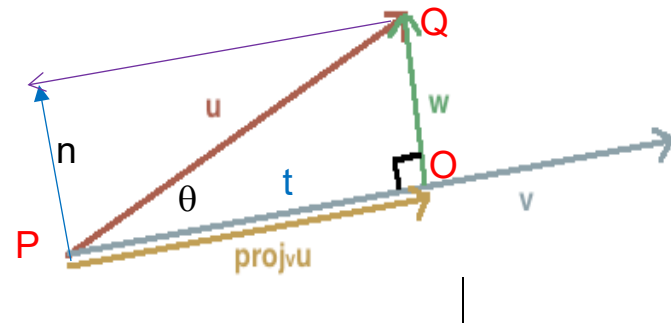


Word Similarity based on Context



- **How to compute similarity between words (vectors)**
 - Dot product
 - Cosine function

Dot (Inner, Scalar) Product



- $u \cdot v = \sum u_i v_i$ (algebraically)
 - $u = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$, $v = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$, $u \cdot v = u^T \cdot v = \sum u_i v_i = 4 * 3 + 5 * 1 = 17$
 - A scalar value that represent the **magnitude** of the **product** of **two vectors**
- $u \cdot v = ||u|| ||v|| \cos \theta$ (geometrically, equivalent to algebraic dot product)
 - From the **projection** of u onto v , $proj_v u = ||u|| \cos \theta$, the **magnitude** of v by the total contribution of u in the direction of v .
 - $\cos \theta = \frac{u \cdot v}{||u|| ||v||}$ (θ can also represents the **proximity** between u and v)
- If u and v are **orthogonal** (perpendicular, $\theta = 90^\circ$, it means independent):
 - $u \cdot v = ||u|| ||v|| \cos \theta = u \cdot v = ||u|| ||v|| * 0 = 0$
- If u and v are **codirectional** (parallel, $\theta = 0^\circ$, it means the most similar):
 - $u \cdot v = ||u|| ||v||$ ($\cos \theta = 1$ when θ is 0°).
 - $u \cdot u = ||u||^2$

Dot Product for Word Similarity



- **Algebraic Dot Product for similarity** between two vectors
 - The algebraic **dot product** represents the **magnitude** (a scalar value) of the product of two vectors.

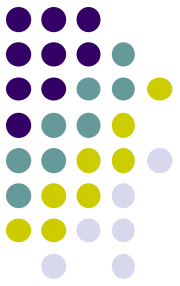
$$u \cdot v = \sum u_i v_i = u_1 v_1 + u_2 v_2 + \cdots + u_N v_N$$

- The dot product tends to be **high** when the two vectors have large values in the **same dimensions** – **useful similarity metric** between vectors.
- **Problem with algebraic dot product**
 - Dot product **favors long vectors** since dot product is higher if a vector is longer (higher values in many dimension).

$$\text{Vector length: } ||v|| = \sqrt{\sum v_i^2}$$

- **Frequent words** (e.g., of, the, you) have **long vectors** (since they occur many times with other words).
 - So, dot product **overly favors frequent words**.

Cosine Similarity



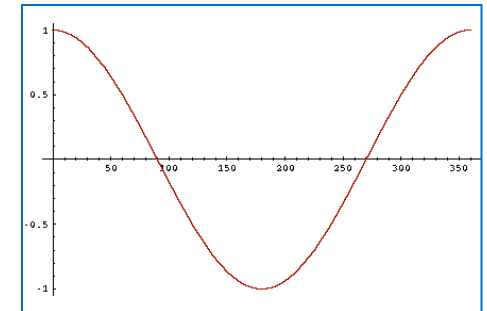
- **Geometric Dot Product for similarity between two vectors**

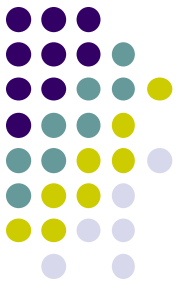
- $u \cdot v = ||u|| ||v|| \cos \theta \Rightarrow \cos \theta = \frac{u \cdot v}{||u|| ||v||}$
- θ between u and v , $\cos(u, v)$ can also represent the **proximity** between u and v .
- $\cos(u, v) = \frac{u \cdot v}{||u|| ||v||}$

- $u \cdot v = \sum u_i v_i = u_1 v_1 + u_2 v_2 + \dots + u_N v_N$ and $||v|| = \sqrt{\sum v_i^2}$

- **Cosine function: $\cos \theta$**

- **+1**: vectors point in same directions
- **-1**: vectors point in opposite directions
- **0**: vectors are **orthogonal**
 - But since **raw frequency values are non-negative**, the **cosine for document vectors** ranges from **0-1**.
 - If we normalize each vector by creating a unit vector of length 1, the dot product is the same as the cosine.





Examples: Word Similarity

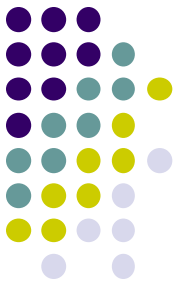
Frequency of words for each document

	pie	data	computer
cherry	442	8	2
digital	5	1683	1670
information	5	3982	3325

$$\cos(u, v) = \frac{u \cdot v}{||u|| ||v||} = \frac{\sum u_i v_i}{\sqrt{\sum u_i^2} \sqrt{\sum v_i^2}}$$

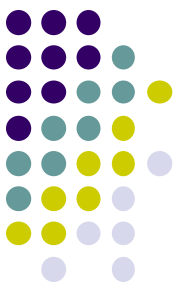
$$\cos(\text{cherry}, \text{information}) = \frac{442 * 5 + 8 * 3982 + 2 * 3325}{\sqrt{442^2 + 8^2 + 2^2} \sqrt{5^2 + 3982^2 + 3325^2}} = .017$$

$$\cos(\text{digital}, \text{information}) = \frac{5 * 5 + 1683 * 3982 + 1670 * 3325}{\sqrt{5^2 + 1683^2 + 1670^2} \sqrt{5^2 + 3982^2 + 3325^2}} = .996$$



Issues: Frequency-based Vectors

- **Word frequencies**
 - Clearly useful, if *sugar* appears a lot near *apricot*, that's useful information but...
- **High dimensionality**
 - The dimension for a unique word, e.g., easily can be 10,000 or more
- **Sparsity**
 - Lots of values may be zeros due to the sparsity of a language
- **Unable to capture semantic relationships**
 - Since the values are frequencies
- **Memory and computational inefficiency**
 - Due to high dimensionality, especially for large vocabularies



Embeddings

- **Word embedding**

- A **dense vector** representation of words that captures **contextual information** about words (semantic relationships) in text data
- **How can we create a dense vector for efficient representation?**
 - Learn co-occurrences of words or related words within a context window, **not every word**
 - Therefore, non-zero in all dimensions
 - Compress the information (by **encoding**)

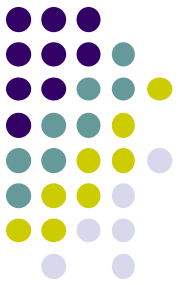
- **Embedding matrix E with dimensions (V, d) : $E = d \times |V|$**

- V : the size of vocabulary; d : the dimensionality of the embedding space (columns); the cell value E_{ij} represents the j -th component of the embedding vector (column) for the i -th word (row).
 - Each row represents the **embedding vector** (or each column if transposed).
 - Each column represents a **dimension** of the embedding space.

- **Capturing the contextual information and semantics**

- **Learning the probabilities** between words.
- **Embedding is a general technique in AI!**
 - Word embeddings, image embeddings, graph embeddings, etc.

Example Word Embedding

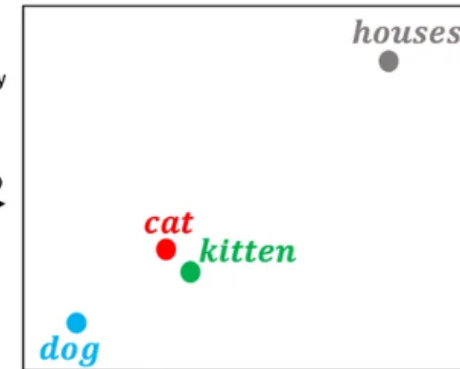


dimension

	living being	feline	human	gender	royalty	verb	plural
<i>cat</i> →	0.6	0.9	0.1	0.4	-0.7	-0.3	-0.2
<i>kitten</i> →	0.5	0.8	-0.1	0.2	-0.6	-0.5	-0.1
<i>dog</i> →	0.7	-0.1	0.4	0.3	-0.4	-0.1	-0.3
<i>houses</i> →	-0.8	-0.4	-0.5	0.1	-0.9	0.3	0.8

Dimensionality
reduction of
word
embeddings
from 7D to 2D

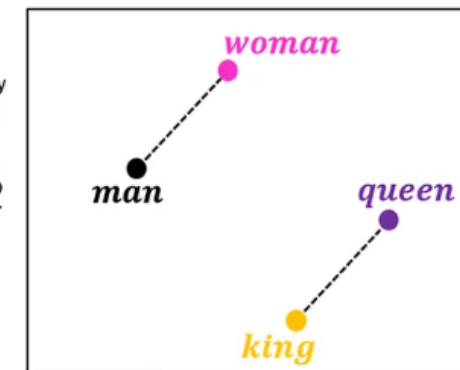
Words in continuous vector space

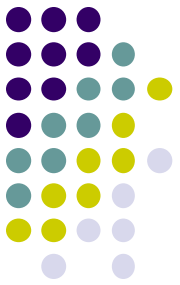


Vectors

<i>man</i> →	0.6	-0.2	0.8	0.9	-0.1	-0.9	-0.7
<i>woman</i> →	0.7	0.3	0.9	-0.7	0.1	-0.5	-0.4
<i>king</i> →	0.5	-0.4	0.7	0.8	0.9	-0.7	-0.6
<i>queen</i> →	0.8	-0.1	0.8	-0.9	0.8	-0.5	-0.9

Dimensionality
reduction of
word
embeddings
from 7D to 2D





Why Dense Vectors?

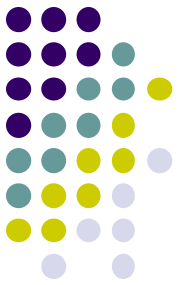
Sparse vectors (TF-IDF, PPMI)

- Long: length $|V| \approx 20,000 - 50,000$
- Sparse: most elements are zero

Dense vectors (embeddings)

- Short: length $|V| \approx 200 - 1000$
- Dense: most elements are non-zero

- **Easier to use as features and less weights to tune**
 - Dense vectors can be created by various methods such as dimensionality reduction or learning from the data through training.
- **Generalize better than count-based vectors**
- **Better at capturing contexts**
- In practice, **they work better!**



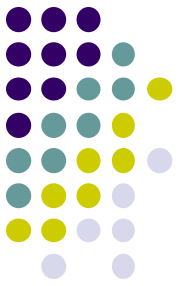
Some Embedding Methods

- **Singular Value Decomposition (SVD)**

- A special case of this is called **Latent Semantic Analysis (LSA)**
 - Captures the underlying structure or patterns in the data – latent semantics
 - **Static embeddings** assign a fixed vector to words

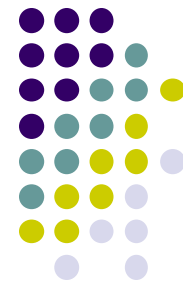
- **Neural Language Models**

- Distributed contextual embeddings **learned through training**
 - **Word2vec**, Glove, FastText (static embedding)
- Distributed contextual embeddings **learned through language modeling**
 - Elmo, BERT, GPT-2, etc. (dynamic embedding)
 - **Dynamic embeddings** assign different vectors to words based on the context (context can change)
- Many more variants
 - Multilingual embeddings, multi-sense embeddings, syntactic embeddings, etc.



Singular Value Decomposition (SVD): Dense Vectors via Latent Space Analysis

Variation and Shape of Distribution



- **Measures of variation** (The differences among values)

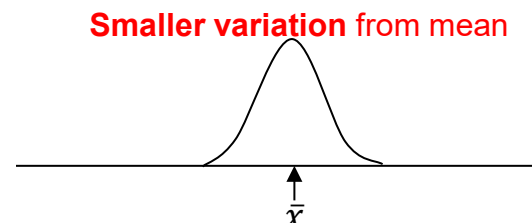
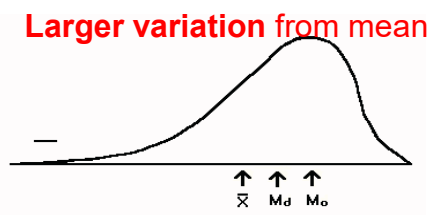
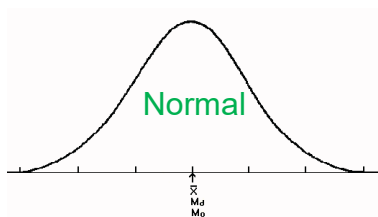
- **Range:** ($x_{\max} - x_{\min}$), **but** sensitive to extreme values
- **Deviation:** ($x - \bar{x}$), **but** the sum of deviation $\sum(x - \bar{x}) = 0$
- **Sample variance:** $s^2 = \frac{\sum(x - \bar{x})^2}{N-1}$ and **standard deviation:** $s = \sqrt{s^2}$
- **Z-score** $= \frac{x - \bar{x}}{s}$

If data is **standardized** (z-score), data is **centered**($\bar{x}=0$), **variance** $s^2=1$, **standard deviation** (SD) $s=1$

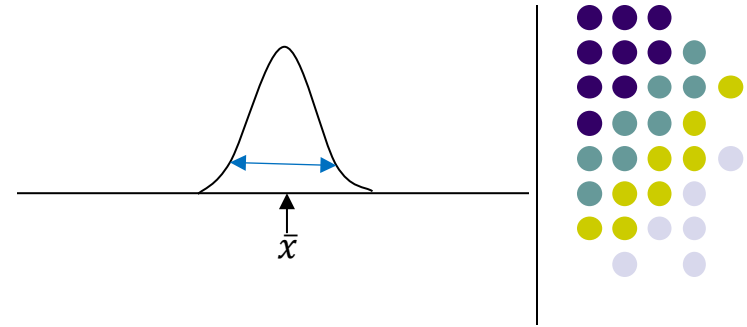
Original data		Centered data		Standardized data	
Body weight (kg)	Body height (cm)	Body weight (kg)	Body height (cm)	Body weight	Body height
86	157	13	-18.33	1.28	-1.46
82	168	9	-7.33	0.89	-0.58
74	170	1	-5.33	0.10	-0.42
73	181	0	5.67	0	0.45
62	191	-11	15.67	-1.08	1.25
61	185	-12	9.67	-1.18	0.77
Mean=73	Mean=175.33	Mean=0	Mean=0	Mean=0	Mean=0
SD=10.16	SD=12.56	SD=10.16	SD=12.56	SD=1	SD=1

- **Shape and variance**

- The differences between the **mean** $\bar{x}$ and values in a data set



Importance of Variance



- **Variance, center value, and data distribution**
 - **Variance** is the measure of the “spread” of data points around their **center** of mass.
 - Groups with different values have different center values.
 - Groups with similar values have similar center values.
 - Groups with little variation in values within themselves, but large variation between them, they would appear to be distinct.
- **Variance is the characteristic of nature and life.**
 - If the variance is zero, the system is deterministic, hence no need for statistics or data analysis since no new information can be obtained (**zero entropy**).
 - **More variance more information it contains.**
 - More variance, increased uncertainty, increased entropy, and more surprise

Variance between Two Variables

- **Covariance**

- A variance of a random variable X_1 , $var(X_1) = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{X}_1)^2$
- A covariance between X_1 and X_2 :

$$cov(X_1, X_2) = \frac{1}{N-1} \sum_{i=1}^N (x_{i1} - \bar{X}_1)(x_{i2} - \bar{X}_2)$$

A measure of the **joint variability** that shows a degree of **strength** and **direction** to which two variables **vary** with respect to each other.

- **Correlation**

- A measure of the **strength** and **direction** of how two variables are **related**.

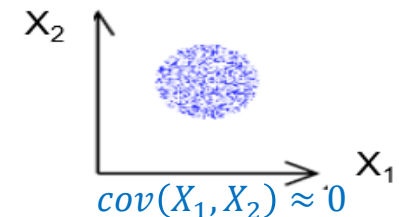
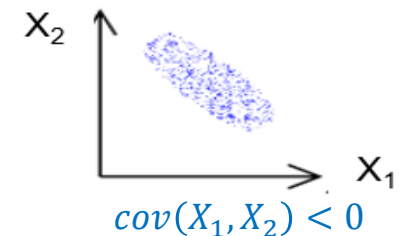
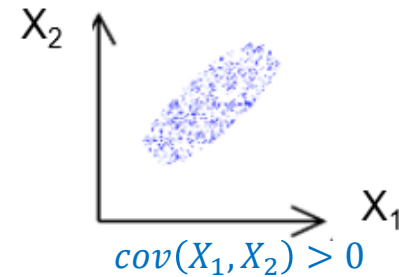
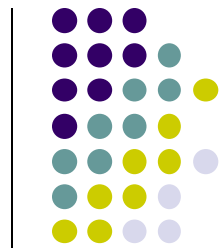
Pearson's correlation coefficient, $r_{X_1 X_2} = \frac{cov(X_1, X_2)}{s_{x1} s_{x2}}$

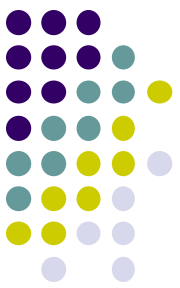
s_{x1} and s_{x2} are standard deviation of variable X_1 and X_2 .

If the **data** are **standardized**, $r_{X_1 X_2} = cov(X_1, X_2)$ since $s_{x1} s_{x2} = 1$.

- **Differences** between covariance and correlation

- **Correlation** is a **standardized** value **[-1,1]** (dimensionless, not affected by the change in scale).
- **Covariance** is **not a standardized** value **[-∞,∞]** (affected by the change in scale and unit).





Covariance Matrix

- **Variance-Covariance matrix**

- **Covariances** are off-diagonal elements for all variable pairs.
- **Variances** are the diagonal elements.

A **covariance matrix** for a data set with **3 features** $\Sigma = \begin{matrix} & \begin{matrix} x_1 & x_2 & x_3 \end{matrix} \\ \begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix} & \begin{bmatrix} Var_1 & Cov_{12} & Cov_{13} \\ Cov_{12} & Var_2 & Cov_{23} \\ Cov_{13} & Cov_{23} & Var_3 \end{bmatrix} \end{matrix}$

- Σ is a **symmetric** and **square matrix**, $\Sigma^T = \Sigma$.
- An element of Σ , Σ_{jm} ($j, m=1, \dots, k$) are computed:

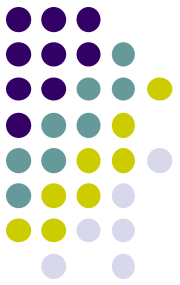
- $\Sigma_{jm}, cov(X_j, X_m) = \frac{1}{N-1} \sum_{i=1}^N (x_{ij} - \bar{X}_j)(x_{im} - \bar{X}_m).$

$$cov(X_j, X_m) = var(X_j, X_j) \text{ when } j=m.$$

- If the data are **standardized** ($\bar{x}=0$, variance $s_j^2=1$), $\Sigma = \frac{1}{N-1} X^T X$

- **Covariance matrix is commonly used in**

- Many mathematical modeling such as **Principal Component Analysis (PCA)**, **Singular Value Decomposition (SVD)**, Gaussian Mixture Models, Kalman filters, etc.



Why Covariance Matrices?

Observation:

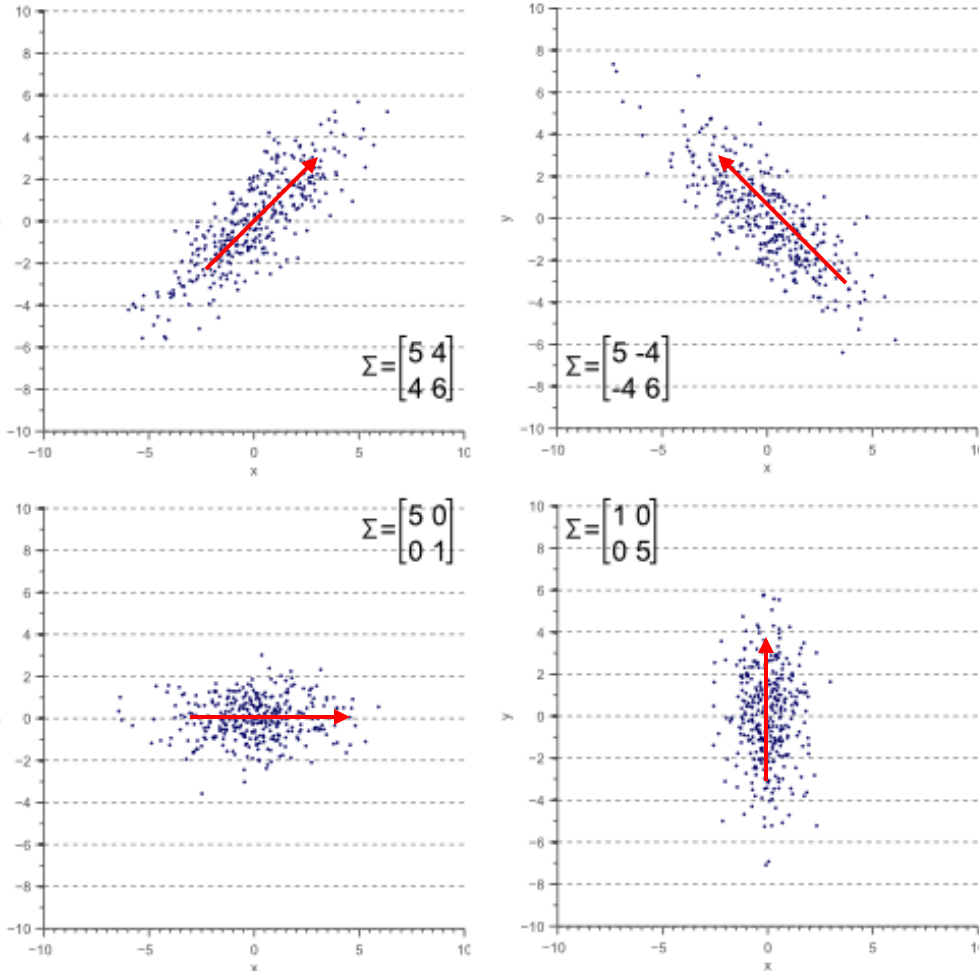
Covariance matrix captures the **shape** of data and **direction of variance**.

Idea:

If we can compute the **primary directions of variance** and ignore the other minor directions of variances,

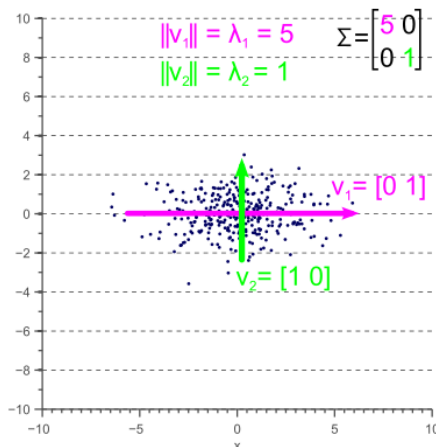
then we may be able to **reduce the dimension** without losing much of **variance (information)** which can also **help** identify important features in a data set.

How can we do that?

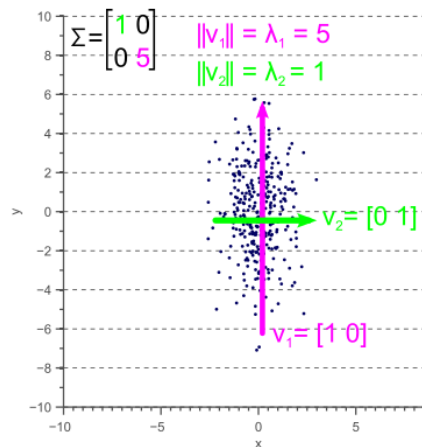


Data distribution and its covariance matrix

Geometric Interpretation of Variance, Eigenvectors, and Eigenvalues



Eigenvectors and eigenvalues of a covariance matrix



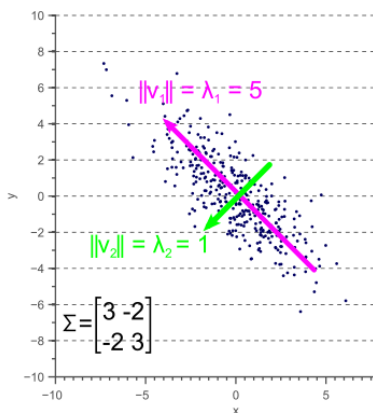
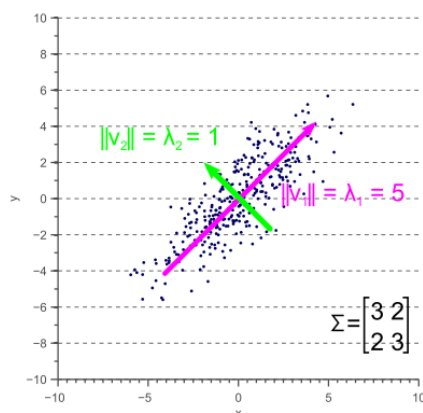
Covariance matrix **captures** the **shape** of data and **direction** of variance in the data set.

$$Av = \lambda v$$

Matrix Eigenvector Eigenvalue

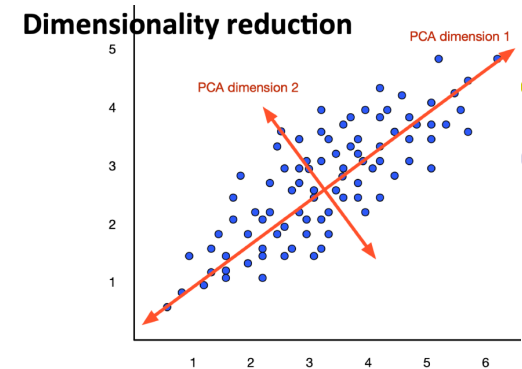
Why eigenvectors and eigenvalues?

- Eigenvectors are **basis vectors** that capture the **inherent patterns** that make up a dataset.
- Eigenvectors of a covariance matrix are the **major directions of variation**.
- Eigenvalues are the **magnitude of the variation** captured by each vector.



Eigenvalues **vs** variance

Dimensionality Reduction



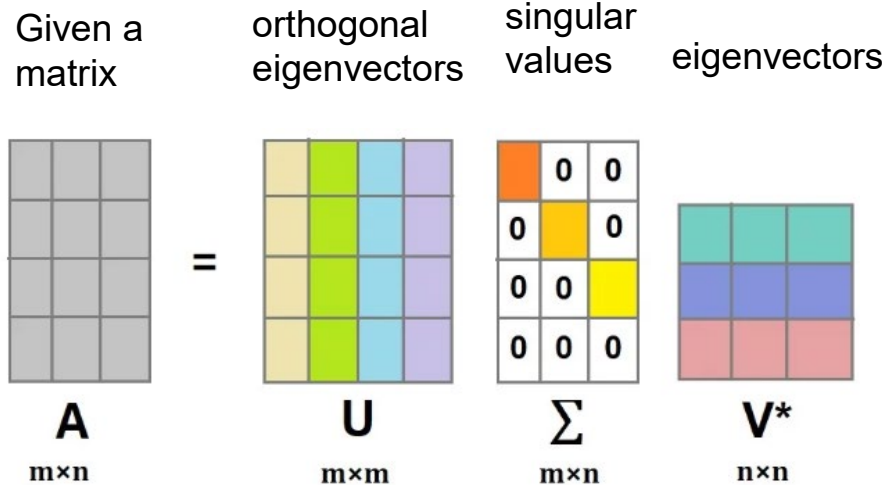
- **Intuition**

- The highest order dimension captures the most variance in the original dataset.
- And the next dimension captures the next highest variance, etc.
- Approximate an N-dimensional dataset to fewer dimensions
 - By first rotating the axes into a new (highly-variant) space (transforming the original dataset into new dimensions by multiplying the data by the selected eigenvectors)

- **Several related methods**

- Principal Component Analysis (PCA)
 - Only works with a covariance matrix (square matrix)
- Singular Value Decomposition (SVD)
 - Generalization of PCA, can work with any matrix, not requires covariance matrix
 - Can be used for Latent Semantic Analysis (LSA)
- Factor Analysis (FA)
 - To find latent factors (a set of latent variables) that explain the covariances

Singular Value Decomposition (SVD)



Given a matrix $A_{m \times n}$, $A = U\Sigma V^T$, U is an orthogonal matrix, whose columns are the **left singular vectors** of A (row space of A)

Σ is a diagonal matrix whose diagonal elements are the (descending) **ordered singular values** of A .

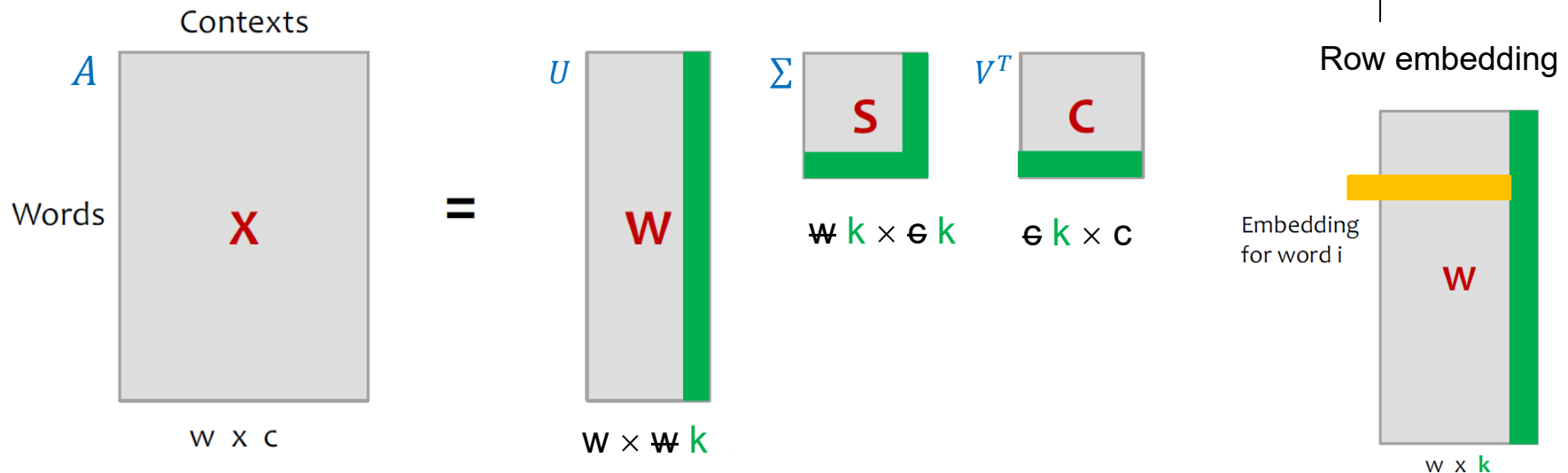
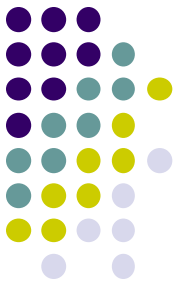
V is an orthogonal matrix, whose columns are the **right singular vectors** of A (column space of A).

- **SVD is a fundamental matrix factorization technique.**

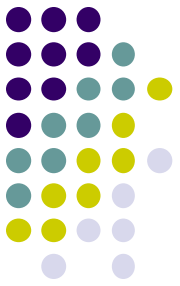
- SVD decomposes A into three parts that describe its core structure.
 - **Rotation** using V^T reorients A to align with the principal directions of variance.
 - **Scaling** using Σ scales the data along with the principal directions found, according to the singular values.
 - **Another rotation** using U rotates the data again to produce the final decomposed matrix.
- The **singular vectors** provide the **direction** of the data.
- The **singular values** representing the **importance** (variance) of each dimension captured in the matrix, arranged in **descending order**.

SVD applied to TDM – Truncated SVD

(for Latent Semantic Analysis)



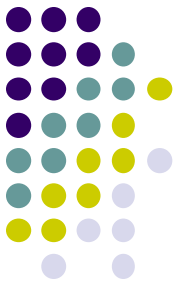
- **Top k largest features to create dense vectors**
 - Keep only the top k largest singular values and their corresponding vectors to get a truncated SVD that helps reduce the dimension but still capture the most important features.
 - Denoising: low-order dimensions may represent unimportant information.
- **Truncated SVD produces word embeddings.**
 - Each row of W is a k -dimensional vector ($k \approx 50-100$) of each word w_i .
 - Generally, we keep the top dimensions, but some experiments suggest removing the top 1 or even 50 dimensions is helpful. Why?



The Brown Clustering Algorithm: A Class-based Word Vectorization Method

The Brown Clustering Algorithm

(Brown et al., 1992)

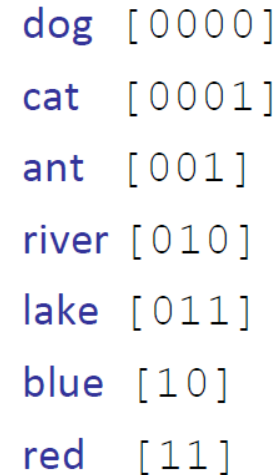


- **Intuition**

- Similar words appear in similar contexts (similar distribution of words to their immediate left and right).
- The algorithm clusters the words based on which the words precede or follow them and produce the output:
 - A partition of words into word clusters or
 - A hierarchical word clusters (a generalization of the clusters)
- These word clusters can be turned into a vector.

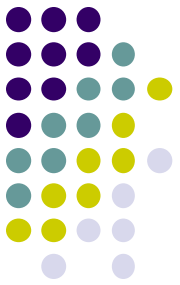
- **Algorithm sketch**

- Each word in a corpus is initially assigned to its own cluster.
- Compute the quality of clusters formed based on similar probabilities of preceding and following the words (mutual information).
- Merge each pair of clusters based on the highest quality of clusters.
- Proceed the clustering until all words are in one big cluster.



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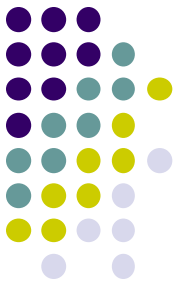
Brown Clustering Example



Friday Monday Thursday Wednesday Tuesday Saturday Sunday weekends Sundays Saturdays
 June March July April January December October November September August
 people guys folks fellows CEOs chaps doubters commies unfortunates blokes
 down backwards ashore sideways southward northward overboard aloft downwards adrift
 water gas coal liquid acid sand carbon steam shale iron
 great big vast sudden mere sheer gigantic lifelong scant colossal
 man woman boy girl lawyer doctor guy farmer teacher citizen
 American Indian European Japanese German African Catholic Israeli Italian Arab
 pressure temperature permeability density porosity stress velocity viscosity gravity tension
 mother wife father son husband brother daughter sister boss uncle
 machine device controller processor CPU printer spindle subsystem compiler plotter
 John George James Bob Robert Paul William Jim David Mike
 anyone someone anybody somebody
 feet miles pounds degrees inches barrels tons acres meters bytes
 director chief professor commissioner commander treasurer founder superintendent dean cus-
 todian
 liberal conservative parliamentary royal progressive Tory provisional separatist federalist PQ
 had hadn't hath would've could've should've must've might've
 asking telling wondering instructing informing kidding reminding bothering thanking deposing
 that tha theat
 head body hands eyes voice arm seat eye hair mouth

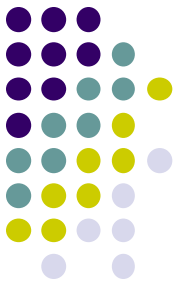
lawyer	1000001101000
newspaperman	100000110100100
stewardess	100000110100101
toxicologist	10000011010011
slang	1000001101010
babysitter	100000110101100
conspirator	1000001101011010
womanizer	1000001101011011
mailman	10000011010111
salesman	100000110110000
bookkeeper	1000001101100010
troubleshooter	10000011011000110
bouncer	10000011011000111
technician	1000001101100100
janitor	1000001101100101
saleswoman	1000001101100110
...	
Nike	1011011100100101011100
Maytag	10110111001001010111010
Generali	10110111001001010111011
Gap	1011011100100101011110
Harley-Davidson	10110111001001010111110
Enfield	101101110010010101111110
genus	101101110010010101111111
Microsoft	10110111001001011000
Ventritex	101101110010010110010
Tractebel	1011011100100101100110
Synopsys	1011011100100101100111
WordPerfect	1011011100100101101000
....	
John	101110010000000000
Consuelo	101110010000000001
Jeffrey	101110010000000010
Kenneth	10111001000000001100
Phillip	101110010000000011010
WILLIAM	101110010000000011011
Timothy	10111001000000001110
Terrence	101110010000000011110
Jerald	101110010000000011111
Harold	1011100100000000100
Frederic	1011100100000000101
Wendell	101110010000000011

Table 1: Sample bit strings



Measuring the Cluster Quality I

- V : vocabulary
- $C: V \rightarrow \{1, 2, \dots, k\}$, a partition of the vocabulary into k clusters
- $e(w_i | C(w_i)) = \frac{\text{count}(w_i)}{\sum_{x \in C(w_i)} \text{count}(x)}$, a probability of word w_i to follow the cluster $C(w_i)$
- $q(C(w_i) | C(w_{i-1}))$, a probability of cluster of w_i to follow the cluster of w_{i-1}
- **The cluster quality:**
 - $\text{Quality}(C) = P(w_1, w_2, \dots, w_n) = \prod_i e(w_i | C(w_i)) q(C(w_i) | C(w_{i-1}))$
 - Or PMI



Measuring the Cluster Quality II

- $Quality(C) = \prod_i e(w_i|C(w_i))q(C(w_i)|C(w_{i-1}))$
- $logQuality(C) = \sum_i \log e(w_i|C(w_i))q(C(w_i)|C(w_{i-1}))$
 $= \sum_{c=1}^k \sum_{c'}^k P(c, c') \log \frac{P(c, c')}{P(c)P(c')} + G$

$$P(c, c') = \frac{n(c, c')}{\sum_{c, c'} n(c, c')} \text{ and } P(c) = \frac{n(c)}{\sum_c n(c)}$$

n : the length of the text. G : a constant

$n(w)$: the # of times word w appears in the text.

$n(w, w')$: the # of times the bigram (w, w') occurs in the text.

$n(c)$: the # of words in a cluster c .

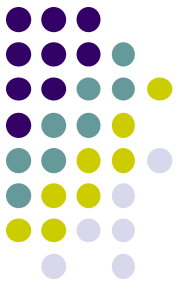
$n(c, c')$: the # of times c' is seen following c , under the function C .

- $Quality(C) = \sum_{c, c'} P(c, c') \log \frac{P(c, c')}{P(c)P(c')} + \sum_w P(w) \log P(w)$
 $= I(C) - H$

$$P(w) = \frac{n(w)}{n}, P(w) = \frac{n(c)}{n}, P(c, c') = \frac{n(c, c')}{n}$$

$I(C)$: mutual information between adjacent clusters

H : the entropy of the word distribution



Pros and Cons of Brown Clustering

- **Pros**

- Unsupervised learning
- Dimensionality reduction
 - By grouping similar words
- Improved features for some NLP tasks
 - The bit-string representation or cluster membership can be used as features in Part-Of-Speech, Named Entity Recognition, etc.

- **Cons**

- Scalability – computationally expensive
- Static clustering